



N Truss Bridge to Emerge from the Effect of Urban Congestion

Adam Youssef – Mokhtar Amr – Yassin Gamal – Youssef Mohammed

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Abstract

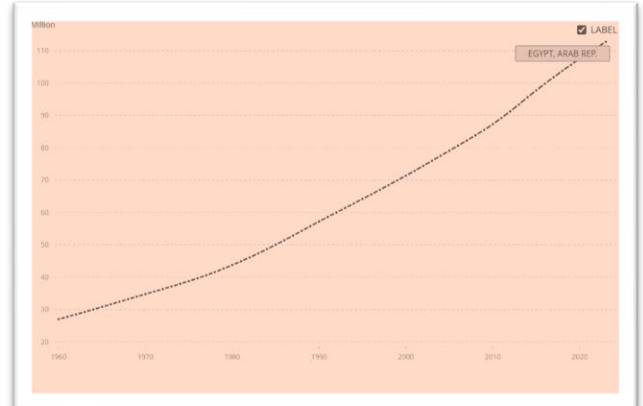
There are eleven major problems that Egypt has suffered throughout the years. Today, they are termed challenges. These challenges inflicted their burden on the Egyptians' well-being and made many of their activities very likely impossible. Some of the challenges sequenced in this essay include towns with nearby populations, a very dry place, and climate change problems that are interrelated and affect one another. Egypt is on the lookout for different palliatives and solutions to this problem. In this regard, there is a made bridge using the N Truss type specifically. The bridge is Vertical-lift. It is a highly efficient mode of operating this bridge and is very reliable. It has a composition with the help of cables and pulleys, used to raise the deck from the ground, allowing boats to float under it. This kind of bridge is easier to design and erect than other bridge types. The price is comparatively cheaper. Also, it requires less maintenance than other similar bridges. Hence, the total cost is less. The Vertical-lift bridge can be operated in different environmental conditions and used according to the need. Weights ranging from 0.5 kg to 10.2 kg would test the performance of the bridge. The maximum deflection noted was 5 mm under the load of 10.2 kg. With no support from the center columns when it was hung in mid-air, there was no deflection under dead load, thereby proving the excellence of the trusses and load distribution.

Introduction

Egypt has faced numerous problems in the preceding years, which can now be termed challenges. Much of such challenges form the major hindrance to the country's development through their effects on national economies and welfare.

Our prototype is expected to affect four major challenges: Urban congestion, population growth, drought, and climate change. Urban congestion is the one that is most targeted in addressing. Urban congestion results in decreased speeds with longer trip times and greater queuing. Those lead to economic impacts, health impacts as well as impacts on the quality of life in the areas that contain it. The major cause of Urban congestion is population growth, which brings about more more-than-usual number of pedestrians and vehicles in the streets, thus leading to traffic jams. It is reported by most demographers in Egypt that the huge population growth is so high. that in the last 6 decades, 70 million individuals have been incorporated into the country's ever-growing population (El Araby, M. (2002) as shown in **Graph 1**

As already mentioned, this increase in population signifies the higher demand from the country for more resources, which is to say that it would affect its economy and welfare since fewer resources will be given to more people. The GCMA, which acts as an economic hub in Egypt, is predicted to house more than 24 million people by 2027, which would impact its productivity and add to urban congestion.



Graph 1: Population growth in Egypt since 1960

Urban congestion is also a result of aridity. Arid areas always undergo quick urbanization through poor road conditions and bad infra structure in arid climates which lead to urban congestion due to migration of the population finding a better job and lifestyle in cities. To curb this urban congestion, Egypt has built numerous cities to ensure proximity to the problems of aridity and overcrowding. El Rhoda Island is an example among such many areas. An area with diversified sectors across economic activity with job opportunities, urban congestion remains the biggest drawback to turning its opportunities into reality, thereby creating inefficiencies, a dip in economic fortunes, and health-related issues. This suggests that there is something more to be done beyond that remedy for the gap, like constructing bridges. The alleviation of congestions is hub specific with building and design specifications. The model for El Rhoda Island bridge's prototype is such that it will have two traffic lanes (7 cm each) and two pedestrian lanes (3 cm each) separated by a 2 cm strip at the center, with a clearance of 10 cm from the lowest point to the water's surface. The bridge should be load-bearing on a minimum of 10 kg with an outright deflection of 5 cm and weighs not more than 20 kg. Research on hydraulic and mechanical motion, Newton's laws, and openable bridge designs fed into the building of the prototype.

Among several advantages, hydraulic motion is easy to adapt with different sizes of ships, as fewer materials are involved, thus very cost effective, very durable against heavy traffic, and even better traffic flow due to reduced waiting times. Specialized maintenance creates a high liability for downtime, but then all hydraulic bridges necessitate this. Among different hydraulic bridge models, the best option for El Rhoda Island would have to be the vertical lift bridge (Shahsavari, V., Mashayekhi, M., Mehrkash, M., & Santini-Bell, E. (2019)). All it needs is a small amount of horizontal space and provides great vertical clearance for vessels passing under it, which makes it most suitable for the 50-meter span proposed, unlike swing bridges.

The longest palm stalk used to build the prototype can't exceed 15 centimetres, with a diameter not greater than 5 mm. Therefore, most parts of the bridge are made by gluing stalks together until the desired length is reached. For instance, in creating the movable (Koglin, T. L. (2003)) part of a bridge which is 28 cm long, 15 cm of palm stalk was placed and complemented by another 13 cm until it reached the required length. The same process followed while constructing the two fixed parts of the bridge, which have a length of about 36 cm each. The placement of palm stalks was carried out in the order of 15 cm, followed by 6 cm, and then another 15 cm, so as to distribute the load over them and minimize as much as possible the deflection. Such characteristics ensure that the bridge is not only effective in eliminating congestion but increasingly effective in traffic flow conversion and even economic productivity-toward formulation of a very significant step towards addressing the major urban challenges facing Egypt.

Materials & Methods

Table 1: Materials and their costs

ITEM	QUANTITY	USAGE	COST	PICTURE
Palm Stalk	150 piece 50 cm each	The basic material for the construction of the bridge.	Recycled	
White Glue	900 g	Gluing Materials together rials together	120 L.E	
Syringe	8	To push the movable part of the bridge.	80 L.E	
Hose Syringe	4 pieces 100 cm each	To link with the syringe and the movable part of the bridge	30 L.E	
Yellow Glue	900 g	Fast gluing material.	50 L.E	

Methods:

The prototype was built in 6 main phases.

Firstly, The palm stalk was cut into 15 cm stalks to be used in the construction As shown in **Figure 1**

Secondly, the base structure was built by the usage of the palm stalk which was used to make a rectangular base then the corners were glued with white glue and the base was reinforced with additional stalk.

Thirdly, the Hinge Mechanism was created through the construction of triangular support at the ends of the base for additional strength. Then cubes were made from palm to create a hinge after that the dowel was cut to size and threaded through the cube to allow the bridge to pivot.

Fourthly, the bridge platform was constructed by gluing sticks in rectangular pattern to form it then the attachment of the platform to the hinge mechanism was ensured so that it can lift and lower smoothly.

Fifthly, the hydraulic system was set up by linking between the syringe and the hose to form the hydraulic system where the input and the output one were acting as switches. Oil was then poured to the syringe to give some pressure. The stable base and the platform were connected to control the lifting mechanism.

Finally, the mechanism was tested and modifications were made to ensure smooth lifting.

Test plan:

Once the prototype is built, an effectiveness test will be conducted, so there shall be tests to determine suitability of the prototype. Deflections, reliability and other proprieties of the prototype are checked by these tests. The prototype deflection test shall be done to find out whether the bridge would be able to support a dead load and a live load with the minimum deflection. In this experiment,

- the centre of the prototype hangs mid-air, where deflections due to dead weight can be observed.

- The midpoint will be calculated by measuring one of the fixed parts' length and getting its half then doing the same process to its width and getting their intersection.

- The midpoint will then be loaded with masses to verify whether it qualifies or fails. Mass will be applied from 0.5 kg to 10 kg incrementing it 0.5 kg in each time.

- For the requirement to be fulfilled, the deflection should not go beyond 5 cm.



Figure 1: Showing the step of cutting

Results

The prototype had a capability for weight support of a maximum of up to 4 kg without showing any significant or considerable additional deflections at all. Beyond the maximum weight, a slight deflection of 1 (±0.5) mm was observed when loaded with a weight of 4.5 kg. Furthermore, the deflection continued gradually as the weight increased adding 1 (±0.5) mm for every several kilograms until it reached a maximum of 7.2 kg. At this point, the amount of deflection increased with an additional load going up 2 (±0.5) mm which indicated that the speed of deflection was two times larger than that which had been previously recorded under regular conditions. The bridge managed to hold over 10.2 kg total weight, much greater than the required mass, under just 5(± 0.5) mm of deflections therefore it is considered successful. As shown in **Table 2**

Table 2: test results

Positive results:

The bridge was able to carry all weights put on it without a significant deflection. It could even carry 0.2 kg more than the required weight and the deflection at its maximum didn't pass 5 mm. This shows how the weight was well distributed on the bridge because of its trusses shape. Also, the bridge's whole mass didn't pass 10 kg which is very low relative to the weights it could carry.

Negative results:

If one of the trusses of the bridge broke or wasn't well maintained it would affect the whole bridge negatively. Also, there is a systematic error in the order of (± 0.5) mm which must be taken into consideration when looking at the performance of the bridge.

Mass (kg)	Deflection (mm)	Weight (N)
0.0	0.0	0.0
0.5	0.0	4.9
1.5	0.0	14.7
2.5	0.0	24.5
3.5	0.0	34.3
4.5	1.0	44.1
5.7	1.0	55.86
6.7	1.0	65.66
7.7	2.0	75.46
8.7	4.5	85.26
9.7	5.0	95.06
10.2	5.0	99.96

Analysis

As shown from the results in the previous section, the prototype showed a variety of positive and negative results but before talking about it, the solution and how it achieved the design requirement should be mentioned. In the prototype,

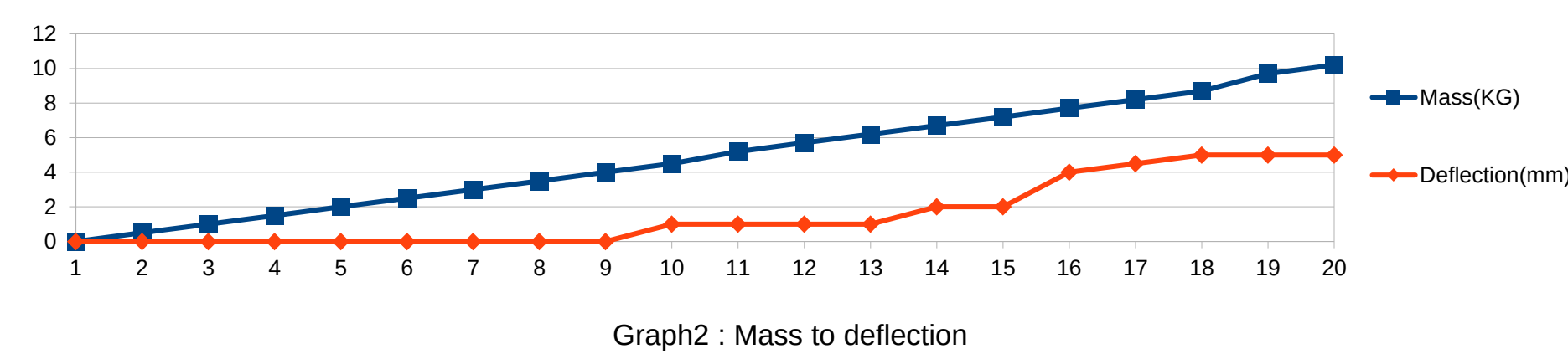
Each one of the fixed parts has the dimensions of 36×22 (±1) cm and the movable part has the dimensions of 28×22 (±1) cm. The height of each fixed part from the ground is 14.5 cm. The four main columns were made by cutting four 15 cm palm stalks and one 10 cm so the combination of them made the height of 55 cm. Most of the trusses in the bridge had lengths varying from 5 cm to 15 cm depending on the place where it would be placed. The height of the movable part is about 24 cm. There were 8 girders made to support the bridge and placed in different parts of it to help the trusses in distributing the deflection. The bridge can support high weights with low deflection. It could carry 10.2 kg with a maximum deflection of 5 mm which is utterly good and shows how the mass is well distributed in the bridge.

The first point is that when mass is increased, the deflection increases at a less rate in comparison with the rest. This goes to show how well-built it is and how it carries the mass over the trusses and columns in the bridge. This is because of the N Truss (Deb, K., & Gulati, S. (2001)), which is the truss that is best in distributing the weights under dynamic load conditions. A feature of this truss is that it has diagonals which form a chain of triangles. These triangles help in supporting live and dead loads. As shown in **Figure 2** In addition,



Figure 2: the bridge during the deflection test

It distributes force over the surfaces of the bridge, which is very important for any bridge because, in this way, force or tension is not able to collapse and fall apart. During the test, these trusses helped to hold the prototype together despite being hanged mid-air and made for very visible results because the weight was distributed throughout every part of the bridge. However, it could carry its dead load because of its structure and the live load added to it during testing. This happens due to the perfect load transfer provided by the geometrical shape, which is the hallmark of the N truss (Krenk, S., Høgsberg, J., Krenk, S., & Høgsberg, J. (2013)). It maintains strength while not wasting the material on it. The N Truss is a truss that handles loads by location so that some members carry compression while other carries tensile loads. The N truss is made to optimize for lightweight materials while being cost-effective. The Pressure was calculated during this test by the law. **Pressure = Force/Area as**



Another reason for involving the N truss design in our prototype as shows in **Figure 3** is that the structure has an outstanding strength-to-weight ratio that could be expressed as how much load the bridge could withstand compared with the weight of the entire bridge itself. The statement may actually be such a huge advantage that one could qualify it as the supreme virtue of an entire truss structure. This is even more true when the total weight of the entire bridge is not more than 20 kilograms



Figure 3: N trusses Shape

From that amazing characteristic, the bridge can sustain a further additional load of 0.2 kilograms from all other comparative bridges tested in the process. In addition, it was discovered that the foundation along with the trusses could carry a weight of about 80 kilograms without much deflection when several alterations were made to the bridge design and structure. Also, another major element influencing the bridge's higher strength-to-weight ratio is rooted in the concepts that **Newton's Third Law of Motion expounds**. This basic law states that **“for every action, there is an equal and opposite reaction.”** As learned in **PH.1.02** In other words, when a load is born by the bridge, every force applied is countered by an equal and opposite force emanating from the opposite direction. Such laws thus prove, with great clarity and conviction, how such a bridge can hold and support several times greater weight than its structure permits. Oddly enough, this is given that there will be such a great and sizeable disparity between the total weight of the bridge and all additional loads it is meant to carry in safety.

And the final reason for its usage is because it has one of the least deflections. The reasons why things deflect are because of the high span length, which leads to greater deflection. They have a relation that **if Span doubles, the deflection increases by 8 times**. Another thing affecting the deflection is the load type because regularly distributed loads have less deflection than concentrated ones As learned in **PH.1.03**. And that's one of the reasons that the maximum length of the palm stalk used in the whole prototype is 15 cm.

The problems that the results highlight are about the case when an irregular loading pattern occurs while the N truss supports a load. At that time, some of the members always get subjected to such a high level of stress, which ultimately results in bending deflection within the structure. This type of construction should be achieved so that, under absolutely proper conditions, weight is uniformly distributed to carry the truss members under idealized conditions. Such perfect construction was, however, never reached during the first trial test of the bridge, even though several revisions and modifications were possible after testing to eliminate the possible errors. The N truss is a construction needing specialists who can feel and measure all angles absolutely to be sure that it is constructed perfectly to serve the purpose.

Secondly, as learned from **PH.1.01**, there was a systematic error of (±0.5) mm because the tape was marked every 1 mm. Such errors are important to be considered because they will affect the prototype as well as the full-scale bridge. This error is quite large when compared to the measured deflections since the error percentage is almost 50% which is a very high percent value and would jeopardize the prototype.

Conclusion

Prototype construction was completed, and the test was conducted. The summarized results are as follows:

- The unloaded height from the ground of the bridge is 14.5 cm.
 - The structure showed no deflections during the tests until weights of up to 4.5 kg were applied, at which point it deflected by 1 mm.
 - The bridge is made from eco-friendly materials and does no harm to the environment.
 - At 10.2 kg, the bridge reached a maximum deflection of 5 mm.
 - The maximum dimension of the truss in the bridge did not exceed 15 cm to ensure that it does not exceed 5 mm.
 - A load of 80 kg can be maintained by the bridge.
- These findings led us to conclude that the bridge meets all design expectations and is therefore considered a success that would help with solving the urban congestion.

Recommendations

However, there could be suggestions that can improve the efficiency and make the prototype more sophisticated in the future:

Including new futuristic technologies that can bring new things into the project. using more recycled materials to make the bridge eco-friendly. Putting as much effort towards the stationary part as the movable part since they are equally essential.

Finding out the best adhesive material and buying it to avoid buying another one in the future which would increase the cost of the prototype.

Proper measurement tools should be used to avoid systematic error.

Suggestions for constructing a large-scale bridge include:

There should be an electricity conducting feature For converting kinetic energy to run mechanisms on it. The materials used should be reliable and elastic capable to withstand earthquakes and deflect without breaking. As shown in **Figure 4**

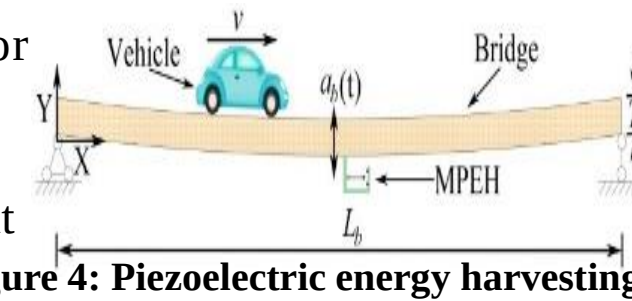


Figure 4: Piezoelectric energy harvesting

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For further information

Name	Email address	Phone number
Adam Youssef	Adam.1724005@stemismailia.moe.edu.eg	01501523533
Mokhtar Amr	Mokhtar.1724042@stemismailia.moe.edu.eg	01148662008
Yassin Gamal	Yassin.1724051@stemismailia.moe.edu.eg	01277213717
Youssef Mohammed	Yousef.1724056@stemismailia.moe.edu.eg	01018086284